An HJB Equation for the Mirrlees Problem: A Structural Parallel with the Ramsey Tax Problem

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Abstract

This note constructs a Hamilton–Jacobi–Bellman (HJB) equation for the Mirrlees optimal nonlinear income-tax problem by exploiting a structural parallel with the continuous-time Ramsey tax problem studied in Jiang, Sargent, and Wang (2025). In the Ramsey problem, decisions are ordered by time $t \geq 0$ and a single implementability constraint attaches a constant Lagrange multiplier Φ_0 to the planner's problem; the value function is linear in a scalar state variable X_t. In the Mirrlees problem, decisions are ordered by the household's productivity type $z \in [z_0, \bar{z}]$ and a single resource constraint attaches a constant multiplier μ; the appropriate HJB has the value function linear in the state variable $\mathcal{U}(z)$—the utility assigned to type z—under quasilinear preferences. In the general case, the HJB still has a canonical form but the value function is not linear in $\mathcal{U}$. The note draws a correspondence table between the two problems.

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1 The Ramsey Problem with No Initial Term Debt

1.1 Environment

Following Jiang, Sargent, and Wang (2025), hereafter JSW, consider a continuous-time Lucas–Stokey economy. A representative household and a benevolent government operate on $t \in [0, \infty)$. The resource constraint is

$$C_t + G_t = N_t, \qquad t \geq 0, \tag{1}$$

where C_t is consumption, N_t is labor supply, and $G_t \geq 0$ is an exogenous government expenditure flow. The household's lifetime utility is

$$\int_0^\infty e^{-\rho t}\, U(C_t, N_t)\, dt, \qquad \rho > 0. \tag{2}$$

Throughout this section we impose the special case:

$$b_{0-,t} = 0 \quad \text{for all } t \geq 0, \tag{3}$$

meaning that the government carries *no* initial term debt.

1.2 The implementability constraint

Using the household's first-order conditions and the government's budget constraint, JSW show that an allocation $\{C_t, N_t\}$ is a competitive equilibrium allocation if and only if it satisfies the resource constraint (1) and the *implementability constraint*. Under (3) this constraint simplifies to

$$\int_0^\infty e^{-\rho t} \big[C_t\, U_{C,t} + N_t\, U_{N,t}\big]\, dt = 0. \tag{4}$$

The Ramsey planner selects the best feasible allocation by maximizing $\int_0^\infty e^{-\rho t} U(C_t, N_t)\, dt$ subject to (1)–(4).

1.3 The cumulative-liability state variable

Because $b_{0-,t} = 0$, the government's cumulative liability Π_t (the net financial position the government would carry if it never restructured its debt) reduces to

$$\Pi_t = \int_0^t \frac{q_{0,s}}{q_{0,t}} \big(G_s - \tau_s N_s\big)\, ds, \qquad \Pi_0 = 0, \tag{5}$$

where $q_{0,t} = e^{-\rho t} U_{C,t}/U_{C,0}$ is the zero-coupon bond price and τ_t is the flat labor-income tax rate. In differential form,

$$d\Pi_t = r_t\, \Pi_t\, dt + (G_t - \tau_t N_t)\, dt, \tag{6}$$

where $r_t = \rho + \dot{C}_t/C_t$ is the equilibrium short rate.

JSW define the scalar state variable

$$X_t = \Pi_t\, U_{C,t}, \tag{7}$$

which encodes both the government's cumulative liability and the household's marginal utility of consumption. Differentiating and using (6) together with the household's first-order conditions, they obtain

$$\dot{X}_t = \rho\, X_t - C_t\, U_{C,t} - N_t\, U_{N,t}, \tag{8}$$

where the $b_{0-,t}$ terms present in the general JSW formula vanish under (3). Note that the dynamics are *linear* in X_t (the ρX_t term).

Initial condition. Because $\Pi_0 = 0$, the state starts at $X_0 = 0$.

1.4 The HJB equation

Define the value function

$$V(X_t, t) = \max_{\{C_s;\, s \geq t\}} \int_t^\infty e^{-\rho(s-t)} U(C_s, C_s + G_s)\, ds, \tag{9}$$

subject to the law of motion (8). By the principle of dynamic programming, V satisfies the HJB equation

$$\rho V = V_t + \max_{C_t}\Big\{U(C_t, C_t + G_t) + V_X\left(\rho X_t - C_t U_{C,t} - N_t U_{N,t}\right)\Big\}. \tag{10}$$

1.5 The linear value-function guess and its verification

Guess. Try the additively separable form

$$V(X_t, t;\, \Phi_0) = -\Phi_0\, X_t + H(t;\, \Phi_0), \tag{11}$$

where $\Phi_0 \geq 0$ is a scalar and H is a function of time and Φ_0. Then $V_X = -\Phi_0$ (constant in both X_t and t).

Verification. Substituting (11) into (10), the $-\Phi_0\,\rho X_t$ terms on both sides cancel, and the HJB reduces to the following ODE for H:

$$\rho\, H(t;\, \Phi_0) = H'(t;\, \Phi_0) + \max_{C_t}\Big\{U(C_t, C_t + G_t) + \Phi_0\big(C_t U_{C,t} + N_t U_{N,t}\big)\Big\}. \tag{12}$$

The maximization over C_t on the right side is independent of the state X_t. The optimal consumption satisfies

$$C(\Phi_0;\, G_t) = \arg\max_{C}\Big\{U(C, C + G_t) + \Phi_0\big(C\, U_C + (C + G_t) U_N\big)\Big\}. \tag{13}$$

1.6 Determining the multiplier Φ_0

Substituting the optimal plan $C^* = C(\Phi_0; G_t)$ and $N^* = C^* + G_t$ into the implementability constraint (4) and imposing $X_0 = 0$, the Ramsey multiplier Φ_0^* is the non-negative root of

$$I_0(\Phi_0) \equiv \int_0^\infty e^{-\rho t}\big[C(\Phi_0;\, G_t)\, U_C + N(\Phi_0;\, G_t)\, U_N\big]\, dt = 0, \tag{14}$$

where $N(\Phi_0; G_t) = C(\Phi_0; G_t) + G_t$.

Remark 1 (Economic interpretation of Φ_0)**.** *The multiplier Φ_0 is the shadow price of the single implementability constraint and is* constant across all dates. *This constancy reflects the exponential discount structure in the Ramsey problem: the ρX_t term in the*

law of motion (8) *exactly cancels the discounting on the left side of the HJB, keeping the shadow price* $-V_X = \Phi_0$ *time-invariant.*

Remark 2 (Constancy of the optimal tax rate when $G_t = G$)**.** *When government spending is constant ($G_t = G$ for all t), the right-hand side of* (13) *does not depend on t, so $C^* = C(\Phi_0^*; G)$ is constant. The implementability constraint* (14) *then collapses to the static condition $C^* U_C^* + N^* U_N^* = 0$, and the Ramsey tax rate is constant at $\tau^* = G/N^*$. This echoes Barro's (1979) tax-smoothing result.*

2 The Mirrlees Problem: Sequential Structure in Productivity

2.1 Environment

Following the lecture notes on optimal nonlinear taxation, there is a continuum of households indexed by productivity $z \in \mathcal{Z} = [z_0, \bar{z}]$ with density $f(z) > 0$. A type-z household produces output

$$y(z) = z\, n(z), \tag{15}$$

where $n(z)$ is labor and z is the marginal product. Utility over consumption c and labor n is $U(c, n)$. The government observes only income y, not the decomposition (z, n).

2.2 The mechanism design formulation

By the taxation principle, the government can equivalently optimize over incentive-compatible (IC) allocations $\{c(z), y(z)\}$. Defining household utility

$$\mathcal{U}(z) = U\left(c(z),\, \frac{y(z)}{z}\right), \tag{16}$$

the local IC condition (envelope theorem applied to the announcement problem) is

$$\frac{d\mathcal{U}}{dz} = -U_n\left(c(z),\, \frac{y(z)}{z}\right) \frac{y(z)}{z^2} \equiv \phi(z), \tag{17}$$

where $\phi(z) > 0$ because $U_n < 0$. Condition (17) is the *envelope ODE*: it plays the role of the law of motion for the state variable when the problem is viewed sequentially in z.

Consumption is the implicit function of state and control:

$$c(z) = C\big(\mathcal{U}(z),\, y(z)\big), \qquad \text{where } \mathcal{U} = U(C(\mathcal{U}, y),\, y/z). \tag{18}$$

Its partial derivatives are $C_{\mathcal{U}} = 1/U_c$ and $C_y = -U_n/(zU_c)$.

2.3 The planner's problem

The government maximizes social welfare subject to the IC law of motion (17) and a resource constraint:

$$\max_{y(\cdot),\mathcal{U}(z_0)} \int_{z_0}^{\bar{z}} \psi(z)\,\mathcal{U}(z)\, f(z)\, dz \tag{19}$$

$$\text{s.t.} \qquad \frac{d\mathcal{U}}{dz} = -U_n\big(C(\mathcal{U}, y),\, y/z\big)\frac{y}{z^2}, \tag{20}$$

$$\int_{z_0}^{\bar{z}} \big[y(z) - C(\mathcal{U}(z), y(z))\big] f(z)\, dz \geq B. \tag{21}$$

The control is the output schedule $y(\cdot)$; the state is $\mathcal{U}(z)$. The initial utility level $\mathcal{U}(z_0)$ is a free endpoint to be optimized.

The single *global* constraint (21) plays the same structural role as the implementability constraint (4) in the Ramsey problem. Attaching a non-negative Lagrange multiplier μ to (21), the augmented objective is

$$\max_{y(\cdot),\mathcal{U}(z_0)} \int_{z_0}^{\bar{z}} \big[\psi(z)\,\mathcal{U}(z) + \mu\big(y(z) - C(\mathcal{U}(z), y(z))\big)\big] f(z)\, dz - \mu B. \tag{22}$$

3 The Mirrlees HJB Equation

3.1 The sequential value function

Viewing productivity z as the "index" along which the planner proceeds (just as t indexes the Ramsey planner's decisions), define the *continuation value* for the planner who has already assigned allocations to types below z and now faces types $[z_0, \ldots] \to$

$[z, \bar{z}]$ with inherited state $\mathcal{U}(z) = u$:

$$V(u, z;\, \mu) = \max_{y(\cdot)} \int_z^{\bar{z}} \big[\psi(\zeta)\,\mathcal{U}(\zeta) + \mu\big(y(\zeta) - C(\mathcal{U}(\zeta), y(\zeta))\big)\big]\, f(\zeta)\, d\zeta, \tag{23}$$

subject to (20) and $\mathcal{U}(z) = u$.

Commitment through promised values. The state variable $u = \mathcal{U}(z)$ is the utility that has already been *promised* to type z by the assignments made to all lower types $\zeta < z$. A continuation planner starting at z is therefore bound to deliver exactly u to type z: it inherits a commitment it cannot revise. In this sense the sequential-in-z formulation embeds full commitment by the original mechanism designer. Types already processed ($\zeta < z$) have received their allocations and cannot be renegotiated; the promised value u is the only "footprint" those past decisions leave on the continuation problem. This is the Mirrlees analogue of the Ramsey planner's commitment to an initial household budget constraint: just as the state variable $X_t = \Pi_t U_{C,t}$ in the Ramsey HJB records the cumulative fiscal obligation that the Ramsey planner cannot escape, the promised utility u in the Mirrlees HJB records the incentive obligation to type z that the Mirrlees planner cannot escape.

3.2 The HJB equation

By the principle of dynamic programming, V satisfies

$$-\frac{\partial V}{\partial z} = \max_y \Big\{ \big[\psi(z)\, u + \mu\big(y - C(u, y)\big)\big]\, f(z) + \frac{\partial V}{\partial u} \cdot \Big(-U_n\big(C(u, y),\, y/z\big)\, \frac{y}{z^2} \Big) \Big\}. \tag{24}$$

This is the *Mirrlees HJB equation.* Writing $\phi(u, y, z) \equiv -U_n(C(u, y), y/z)\, y/z^2 > 0$, (24) can be compactly stated as

$$-V_z = \max_y \Big\{ [\psi(z)\, u + \mu(y - C(u, y))] f(z) + V_u \cdot \phi(u, y, z) \Big\}. \tag{25}$$

Boundary condition. At the top of the type distribution there is no continuation value, so

$$V(u, \bar{z};\, \mu) = 0 \quad \text{for all } u. \tag{26}$$

3.3 The linear value-function guess

Guess. Motivated by the Ramsey analogy, try

$$V(u, z;\, \mu) = \lambda(z;\, \mu) \cdot u + K(z;\, \mu), \tag{27}$$

where λ is a function of z and K is a separate function of z and μ. Then $V_u = \lambda(z)$.

When does the guess work? The linear guess is valid whenever the maximand in (25) is *affine in* u. This requires:

(i) The flow payoff $[\psi(z)u + \mu(y - C(u, y))]f(z)$ to be affine in u, which holds when $C(u, y)$ is affine in u.

(ii) The state dynamics $\phi(u, y, z)$ to be independent of u (or affine in u), which holds when $U_n(C(u, y), y/z)$ is independent of u.

Both conditions are satisfied under *quasilinear preferences*, $U(c, n) = c - v(n)$, since then $C(u, y) = u + v(y/z)$, which is linear in u with slope 1, and $U_n = -v'(y/z)$ is independent of c and hence of u.

Verification under quasilinear preferences. With $U = c - v(n)$:

$$\begin{aligned} C(u, y) &= u + v(y/z), \\ \phi(u, y, z) &= v'(y/z)\, y/z^2. \end{aligned}$$

Substituting the linear guess (27) into (25):

$$\begin{aligned} &-\lambda'(z)\, u - K'(z) \\ &\qquad = \max_y \Big\{ \big[\psi(z) - \mu\big] u\, f(z) + \mu\big[y - v(y/z)\big] f(z) + \lambda(z)\, v'(y/z)\, y/z^2 \Big\}. \end{aligned}$$

Separating terms in u from terms independent of u:

$$\lambda'(z) = \big[\mu - \psi(z)\big] f(z), \tag{28}$$

$$K'(z) = -\max_y \Big\{ \mu\big[y - v(y/z)\big] f(z) + \lambda(z)\, v'(y/z)\, y/z^2 \Big\}. \tag{29}$$

Equation (28) is exactly the costate equation derived from the Hamiltonian approach in the lecture notes.

3.4 Optimality condition for the output schedule

The first-order condition for the maximization problem in (29) (in the quasilinear case) is

$$0 = \mu\big[1 - v'(y/z)/z\big] f(z) + \frac{\lambda(z)}{z^2}\big[v'(y/z) + n\, v''(y/z)\big], \tag{30}$$

where $n = y/z$. Rearranging using $v'(n) = z(1 - T'(y))$:

$$\frac{T'(y(z))}{1 - T'(y(z))} = \left[1 + \frac{n\, v''(n)}{v'(n)}\right] \cdot \frac{\lambda(z)}{z\, f(z)\, \mu}. \tag{31}$$

This is the Diamond (1998) ABC formula.

3.5 Boundary conditions and determination of μ

At the top. The boundary condition (26) with the guess (27) gives

$$\lambda(\bar{z}) = 0, \qquad K(\bar{z}) = 0. \tag{32}$$

At the bottom. The initial utility $\mathcal{U}(z_0)$ is a free endpoint (the planner can choose the utility of the lowest type). The transversality condition at the free endpoint is $V_u(u, z_0) = 0$, which by the guess (27) gives

$$\lambda(z_0) = 0. \tag{33}$$

Together, (32)–(33) are the two-point boundary conditions on the costate ODE (28). They are exactly the transversality conditions $\lambda(z_0) = \lambda(\bar{z}) = 0$ derived in the lecture notes. In the quasilinear case, integrating (28) from z to $\bar{z}$ using $\lambda(\bar{z}) = 0$ gives

$$\lambda(z) = \int_z^{\bar{z}} \big[\psi(\zeta) - \mu\big] f(\zeta)\, d\zeta, \tag{34}$$

and the lower boundary condition $\lambda(z_0) = 0$ pins down $\mu = 1$ (the marginal utility of public funds equals one under quasilinear preferences).

Determining μ in general. For general preferences, the multiplier μ is determined by the resource constraint (21):

$$\int_{z_0}^{\bar{z}} \big[y^*(z) - C(\mathcal{U}^*(z), y^*(z))\big] f(z)\, dz = B, \tag{35}$$

which plays the role of equation (14) in the Ramsey problem.

4 Structural Correspondence between the Two Problems

Table 1 displays the element-by-element structural parallel between the Ramsey problem (no initial term debt) and the Mirrlees problem.

A key structural difference deserves emphasis:

Remark 3 (Constant vs. varying shadow price)**.** *In the Ramsey problem, the shadow price $V_X = -\Phi_0$ is the same constant at every date t. This constancy arises because the state dynamics* (8) *contain a ρX_t term that cancels the discounting ρV on the left of the HJB* (10)*, leaving the costate equation trivially satisfied with $\Phi_0 = const$.*

In the Mirrlees problem, the shadow price $\lambda(z) = V_{\mathcal{U}}$ evolves with z according to the costate ODE (28)*. The reason is that the envelope ODE* (17) *contains no term proportional to $\mathcal{U}$ (analogous to no "z-discounting"), and the welfare weight $\psi(z)$ varies with z. The costate $\lambda(z)$ accumulates the gap between the social welfare weight ψ and the social cost of public funds μ, measuring the "redistribution pressure" emanating from types above z.*

Remark 4 (Quasilinear preferences as the exact Ramsey analogue)**.** *In the Ramsey problem, the linear value-function guess $V = -\Phi_0 X_t + H(t)$ works for* all *admissible preferences. This is because the dynamics* (8) *are linear in X_t for any preferences, and the optimization over C_t in the HJB is independent of X_t.*

In the Mirrlees problem, the analogous linear guess $V = \lambda(z)\mathcal{U} + K(z)$ works only under quasilinear preferences *$U = c - v(n)$, because only then is the implicit function $C(\mathcal{U}, y) = \mathcal{U} + v(y/z)$ linear in $\mathcal{U}$, making the HJB affine in the state $\mathcal{U}$. For general preferences, $C(\mathcal{U}, y)$ is nonlinear in $\mathcal{U}$, and the value function $V(u, z)$ is nonlinear in u. The HJB* (25) *remains the correct dynamic-programming equation in both cases; it is only the validity of the* linear *guess that is special to quasilinear preferences.*

5 The Mirrlees HJB in the General Case

For general preferences, the HJB (25) still governs the value function, but V is a nonlinear function of u. The optimality conditions derived from the Hamiltonian approach

Table 1: Structural Correspondence: Ramsey vs. Mirrlees

Ramsey problem (no initial debt)	**Mirrlees problem**
Sequential index: time $t \in [0, \infty)$	Sequential index: productivity $z \in [z_0, \bar{z}]$
State variable: $X_t = \Pi_t U_{C,t}$	State variable: $\mathcal{U}(z)$ (household utility)
Control: consumption C_t	Control: output schedule $y(z)$
Law of motion: $\dot{X}_t = \rho X_t - C_t U_{C,t} - N_t U_{N,t}$	Envelope ODE: $\mathcal{U}'(z) = -U_n(c, y/z)\, y/z^2$
Initial condition: $X_0 = 0$	Free endpoint: $\mathcal{U}(z_0)$ optimized; $\lambda(z_0) = 0$
Terminal condition: transversality as $t \to \infty$	Terminal condition: $\lambda(\bar{z}) = 0$
Objective: $\int_0^\infty e^{-\rho t} U(C_t, N_t)\, dt$	Objective: $\int_{z_0}^{\bar{z}} \psi(z)\, \mathcal{U}(z)\, f(z)\, dz$
Single global constraint: implementability $(= 0)$	Single global constraint: resource $(= B)$
Global-constraint multiplier Φ_0	Global-constraint multiplier μ
Value function: $V = -\Phi_0 X_t + H(t; \Phi_0)$	Value function: $V = \lambda(z)\mathcal{U} + K(z; \mu)$ (linear in $\mathcal{U}$ under quasilinear preferences)
HJB shadow price: $V_X = -\Phi_0$ *(constant)*	HJB shadow price: $V_{\mathcal{U}} = \lambda(z)$ *(varies with z)*
Costate dynamics: $-\Phi_0 = \text{const}$	Costate ODE: $\lambda'(z) = [\mu - \psi(z)] f(z)$
Optimal control: $C(\Phi_0; G_t) = \arg\max_C [\ldots]$	Optimal control: $y(z) = \arg\max_y [\ldots]$ (tax formula)
Determines Φ_0: $I_0(\Phi_0) = 0$	Determines μ: resource constraint $= B$

in the lecture notes—the $H_y = 0$ condition and the costate equation—are exactly the first-order conditions of the HJB: they arise from differentiating the maximand in (25) with respect to y and equating the z-derivative of $V_{\mathcal{U}}$ to minus the $\mathcal{U}$-gradient of the maximand.

Specifically, denote $\lambda(z) \equiv V_u(u^*(z), z)$ along the optimal path, where $u^*(z) = \mathcal{U}^*(z)$ is the optimal utility assignment. Then:

(i) **Optimality in** y (from $\partial/\partial y$ of the maximand):

$$\mu\,[1 - C_y]f(z) - \lambda(z)\Big[U_{nc}\,C_y + U_{nn}/z\Big]\frac{y}{z^2} - \lambda(z)\frac{U_n}{z^2} = 0. \tag{36}$$

(ii) **Costate equation** (from $-\partial/\partial u$ of the maximand):

$$\frac{d\lambda}{dz} = \mu\,C_{\mathcal{U}}\,f(z) - \psi(z)\,f(z) + \lambda(z)\,U_{nc}\,\frac{y}{z^2}\,C_{\mathcal{U}}. \tag{37}$$

These are identical to equations (5.2) and (5.3) of the lecture notes. The HJB perspective makes clear that $\lambda(z)$ is not an auxiliary variable of the Hamiltonian approach but is the *slope of the value function* $\lambda(z) = V_u(\mathcal{U}^*(z), z)$ evaluated along the optimal path.

6 Summary and Interpretation

Both the Ramsey and Mirrlees planning problems share the following structural architecture:

(1) **A sequential ordering principle.** In Ramsey, decisions are made along the time axis t. In Mirrlees, the social planner processes households from $z = z_0$ upward through higher productivities. In both cases, the planner integrates a *flow payoff* and a *law of motion* for a one-dimensional state variable.

(2) **A single global constraint that attaches a constant multiplier.** In Ramsey, the implementability constraint (4) attaches the constant Φ_0. In Mirrlees, the resource constraint (21) attaches the constant μ. In both cases, this multiplier folds the global constraint into the flow payoff and turns the problem into a standard (undiscounted or discounted) calculus-of-variations problem.

(3) **A Hamilton–Jacobi–Bellman equation.** Once the global constraint is Lagrangianized, the problem admits a value function that satisfies an HJB. The shadow price of the state variable ($V_X = -\Phi_0$ in Ramsey; $V_{\mathcal{U}} = \lambda(z)$ in Mirrlees) is the costate variable of the underlying optimal control problem.

(4) **A determining equation for the global multiplier.** The multiplier is found by imposing the global constraint: $I_0(\Phi_0) = 0$ in Ramsey, and the resource constraint (35) in Mirrlees.

The deepest difference is that in Ramsey the costate is constant (because exponential discounting with rate ρ exactly offsets the ρX_t drift in the state), while in Mirrlees the costate $\lambda(z)$ varies, accumulating the difference between the planner's welfare weights and the social cost of funds. Under quasilinear preferences the linear value-function guess works in both problems, and the two HJBs are fully isomorphic.

7 Extension: Cyclical Government Expenditures in the Ramsey Problem

7.1 Setup

This section studies the Ramsey HJB under cyclical government expenditures, mirroring Section 5.3 of JSW. Maintain the no-initial-debt specialization $b_{0-,t} = 0$ and adopt the utility function $U(C, N) = \ln C - N$ used in JSW's quantitative examples. Government spending follows

$$G_t = G_0 + \theta_G \cos(\omega t), \qquad G_0,\, \theta_G > 0,\ \omega > 0, \tag{38}$$

so that fiscal needs oscillate around the mean G_0 with amplitude θ_G and angular frequency ω.

7.2 Constant optimal tax from the HJB

Under $U = \ln C - N$ we have $U_C = 1/C$ and $U_N = -1$. With $b_{0-,t} = 0$, the maximization problem (13) in the HJB becomes

$$C(\Phi_0;\, G_t) = \arg\max_{C>0}\Big\{\ln C - (C+G_t) + \Phi_0\big[C\cdot\tfrac{1}{C} + (C+G_t)(-1)\big]\Big\} = \arg\max_{C>0}\Big\{\ln C - (1+\Phi_0)(C+G_t)\Big\}. \tag{39}$$

The first-order condition $1/C = 1 + \Phi_0$ yields

$$C^* = \frac{1}{1+\Phi_0^*}, \qquad \tau^* = 1 - C^* = \frac{\Phi_0^*}{1+\Phi_0^*}. \tag{40}$$

Crucially, C^* and τ^* are *independent of* G_t: the cyclical component of fiscal needs does not appear in the static optimality condition for consumption. The Ramsey planner therefore imposes a *constant* tax rate on every date, even though government expenditures oscillate. This is the HJB incarnation of Barro's (1979) tax-smoothing result.

Remark 5 (Why G_t drops out)**.** *The HJB maximand contains G_t twice: once in the felicity term $-(C + G_t)$ and once in the incentive term $-(1 + \Phi_0)G_t$. Both enter additively as constants in the maximization over C. The optimum C^* is therefore pinned by the ratio of the marginal utility of consumption to the Lagrange multiplier, with no role for G_t.*

7.3 Determining Φ_0^* from the implementability constraint

The implementability constraint (14) with $U_C = 1/C^*$ and $U_N = -1$ reduces to

$$\int_0^\infty e^{-\rho t}\big[1 - (C^* + G_t)\big]\, dt = 0, \tag{41}$$

which gives

$$\frac{1 - C^*}{\rho} = \int_0^\infty e^{-\rho t} G_t\, dt = \frac{G_0}{\rho} + \theta_G \frac{\rho}{\rho^2 + \omega^2}. \tag{42}$$

Hence

$$1 - C^* = G_0 + \theta_G \frac{\rho^2}{\rho^2 + \omega^2}. \tag{43}$$

Three observations follow. First, the equilibrium tax rate depends only on the discounted present value of government expenditures, not on the timing of the oscilla-

tions. Second, the contribution of the cyclical component is attenuated by the factor $\rho^2/(\rho^2+\omega^2) \in (0,1)$: high-frequency fiscal cycles ($\omega \gg \rho$) contribute negligibly to the tax rate because they nearly cancel in present value. Third, as $\theta_G \to 0$ the formula collapses to the constant-spending result $1 - C^* = G_0$.

7.4 The cyclical state variable X_t^*

Even though C^* and τ^* are constant, the state variable $X_t^* = \Pi_t^* U_{C,t}^* = \Pi_t^*/C^*$ inherits the cyclicality of G_t. Substituting $C^* = \text{const}$, $N_t^* = C^* + G_t$ into the law of motion (8):

$$\dot{X}_t = \rho X_t - 1 + C^* + G_t = \rho X_t - \frac{\theta_G \rho^2}{\rho^2+\omega^2} + \theta_G \cos(\omega t), \tag{44}$$

where we used $1 - C^* = G_0 + \theta_G \rho^2/(\rho^2+\omega^2)$ to absorb G_0. The ODE (44) is a linear constant-coefficient equation with a sinusoidal forcing term and a constant term that exactly offsets the particular solution's mean. Solving with $X_0 = 0$:

$$X_t^* = \frac{\theta_G}{\rho^2+\omega^2}\big[\rho\big(1-\cos(\omega t)\big) + \omega \sin(\omega t)\big]. \tag{45}$$

This solution is bounded, oscillates with period $2\pi/\omega$, and starts and returns to values near zero. For $\omega \gg \rho$ the amplitude of oscillation is approximately θ_G/ω, so high-frequency spending shocks produce only small oscillations in the state variable.

Remark 6 (X_t^* as a fiscal buffer)**.** *The state variable X_t^* is the product of the government's cumulative liability Π_t^* and the household's marginal utility $1/C^*$. Under constant C^*, variations in X_t^* map one-for-one into variations in Π_t^*. Equation* (45) *shows that Π_t^* absorbs all cyclical fiscal fluctuations: when G_t is high the government runs a deficit (the primary surplus $S_t^* = \tau^* N_t^* - G_t$ falls), pushing Π_t^* up; when G_t returns to its mean, the surplus rises and Π_t^* falls back. The tax rate is constant throughout because it is the* state variable*, not the tax rate, that adjusts.*

7.5 Structure of $H(t;\, \Phi_0^*)$

The time-varying component of the value function, $H(t;\, \Phi_0^*)$, satisfies the ODE (12) with $C_t = C^*$ fixed. Evaluating the maximand:

$$\rho H(t;\, \Phi_0^*) = H'(t;\, \Phi_0^*) + \ln C^* - (C^* + G_t) + \Phi_0^*\big[1 - (C^* + G_t)\big]. \tag{46}$$

Because $G_t = G_0 + \theta_G \cos(\omega t)$, the right-hand side oscillates in t. The solution $H(t;\, \Phi_0^*)$ is therefore a sum of a constant steady-state component (driven by G_0) and an oscillatory transient (driven by $\theta_G \cos(\omega t)$):

$$H(t;\, \Phi_0^*) = H_\infty + A_H \cos(\omega t) + B_H \sin(\omega t), \tag{47}$$

where $H_\infty = [\ln C^* - (1 + \Phi_0^*)C^*]/\rho$ is the constant component and (A_H, B_H) are determined by matching ω-frequency terms. Although H oscillates, the optimal C does not: the oscillations in H represent the time-varying cost of financing cyclical government expenditures, holding the tax plan fixed. The value function $V(X_t^*, t) = -\Phi_0^* X_t^* + H(t; \Phi_0^*)$ therefore oscillates both through the state X_t^* and directly through $H(t)$, the two oscillations together reflecting the complete welfare cost of the cyclical fiscal program.

Remark 7 (Comparison with the constant-G case)**.** *When $\theta_G = 0$ the problem is fully stationary: $C^* = 1/(1 + \Phi_0^*)$, $\tau^* = G_0/N^* = G_0/(C^* + G_0)$ is constant, $r^* = \rho$ is constant, $X_t^* = 0$ for all t, and $H(t; \Phi_0^*) = H_\infty$ is constant. The HJB collapses to a static saddle-point problem. Introducing $\theta_G > 0$ breaks stationarity of H and X^* but leaves the control rule intact: the optimal plan "reacts" to fiscal cycles entirely through the state variable, not through the tax rate.*

8 Extension: The Mirrlees HJB with Pareto-Distributed Productivity

8.1 Setup and motivation

The canonical Mirrlees analysis with bounded support $\mathcal{Z} = [z_0, \bar{z}]$ yields a *zero top marginal tax rate*: $T'(y(\bar{z})) = 0$. This result follows directly from the boundary condition $\lambda(\bar{z}) = 0$ together with the tax formula (31): at the top, $(1 - F(\bar{z})) = 0$ so the costate is zero and the distortion vanishes. A large empirical literature documents that the upper tail of the earnings distribution is well-approximated by a Pareto distribution, not a distribution with bounded support. Diamond and Saez (2011) argue that the zero-top-rate result is an artifact of bounded support and derive a formula for the optimal *limiting* marginal tax rate under an unbounded Pareto tail.

This section extends the HJB framework to unbounded support $\mathcal{Z} = [z_0, \infty)$ and derives the Diamond–Saez top-rate formula directly from the asymptotic behavior of the Mirrlees HJB.

8.2 The Pareto distribution

Let productivity z follow a Pareto distribution with shape $\alpha > 1$ and lower bound z_0:

$$f(z) = \alpha\, z_0^{\alpha}\, z^{-\alpha-1}, \qquad 1 - F(z) = \left(\frac{z_0}{z}\right)^{\alpha}, \qquad z \geq z_0. \tag{48}$$

The inverse hazard rate is

$$\frac{1 - F(z)}{z\, f(z)} = \frac{1}{\alpha} \quad \text{for all } z \geq z_0, \tag{49}$$

a constant. Under the optimal tax schedule, earnings $y(z)$ are also Pareto in the tail. With quasilinear preferences $U = c - v(n)$ and constant elasticity $v(n) = n^{1+1/e}/(1 + 1/e)$, the earnings Pareto parameter is

$$\alpha_y = \frac{\alpha\, e}{1 + e}, \tag{50}$$

where $e = 1/\gamma$ is the compensated labor-supply (or taxable-income) elasticity.

8.3 Transversality replaces the upper boundary condition

With $\bar{z} = \infty$, there is no upper type to impose $\lambda(\infty) = 0$ as a point condition. Instead, optimality requires the transversality condition

$$\lim_{z \to \infty} V(u,\, z;\, \mu) = 0 \quad \text{for all } u, \tag{51}$$

meaning no continuation value remains once all types have been assigned allocations. Under the linear guess $V = \lambda(z)\, u + K(z; \mu)$, this decomposes into

$$\lambda(\infty) = 0, \qquad K(\infty) = 0. \tag{52}$$

These are exactly the conditions that ensure the costate integral (34) converges. For the Pareto distribution, the integrand $[\psi(\zeta) - \mu] f(\zeta)$ decays as $\zeta^{-\alpha-1}$, fast enough for $\alpha > 1$, so the integral converges and $\lambda(\infty) = 0$ is automatically satisfied as a

transversality condition rather than imposed as a point constraint.

8.4 Asymptotic costate and the nonzero top rate

Specialize to quasilinear preferences ($\mu = 1$) and assume the planner places zero welfare weight on the top: $\psi(z) \to 0$ as $z \to \infty$ (a purely redistributive planner). For z large enough that $\psi(z) = 0$:

$$\lambda(z) = \int_z^\infty \big[\psi(\zeta) - 1\big] f(\zeta)\, d\zeta \;\approx\; -\int_z^\infty f(\zeta)\, d\zeta = -(1 - F(z)) = -\Big(\frac{z_0}{z}\Big)^\alpha < 0. \quad (53)$$

The costate is *negative* for high types and decays to zero at the Pareto rate $z^{-\alpha}$. Negative $\lambda(z)$ means the value function V is *decreasing* in the inherited utility $\mathcal{U}$: giving more utility to a high-productivity type is socially costly because it tightens the incentive constraint for all lower types, requiring the planner to give them more utility to maintain IC.

Remark 8 (Negative costate and redistribution)**.** *In the bounded-support problem, $\lambda(z) \geq 0$ everywhere under redistributive weights (the costate equals the redistribution pressure from above, which is non-negative). With Pareto unbounded support and zero tail weight, $\lambda(z) < 0$ for z above the crossing point $\psi(z^{**}) = 1$. The sign reversal signals that the planner is willing to earn revenue from top types at the cost of distorting their labor supply—exactly what a positive top marginal tax rate achieves.*

8.5 The Diamond–Saez formula from the asymptotic HJB

Substituting the asymptotic costate (53) into the tax formula derived from the HJB first-order condition (the general formula (31) applied to the quasilinear case with constant elasticity $e = 1/\gamma$):

$$\begin{aligned} \frac{T'}{1 - T'} &= -\frac{1 + \varepsilon^u}{\varepsilon^c} \cdot \frac{\lambda(z)}{z\, f(z)\, \mu} \\ &\approx \frac{1 + \gamma}{} \cdot \frac{1 - F(z)}{z\, f(z)} \\ &= \frac{1 + \gamma}{\alpha}, \end{aligned} \quad (54)$$

where the second line uses $\lambda(z)/\mu \approx -(1 - F(z))$ and the third uses the Pareto inverse hazard rate (49). Since $(1+\varepsilon^u)/\varepsilon^c = (1+1/e)/(1/e) = 1+\gamma$ for quasilinear preferences,

and converting to the earnings Pareto parameter via $\alpha = \alpha_y(1+e)/e = \alpha_y(1+\gamma)$:

$$\frac{T'_\infty}{1-T'_\infty} = \frac{1+\gamma}{\alpha_y(1+\gamma)} = \frac{1}{\alpha_y\, e}, \tag{55}$$

giving the Diamond–Saez formula:

$$\boxed{T'_\infty = \frac{1}{1+\alpha_y\, e}.} \tag{56}$$

8.6 Interpretation through the HJB lens

Formula (56) can be read directly from three elements of the HJB:

(i) **Costate decay rate** (α)**.** $-\lambda(z)/[1-F(z)] \to 1$: the costate decays at exactly the rate of the Pareto tail. A heavier tail (α smaller) means the costate $\lambda(z)$ is numerically large relative to density, so the incentive-constraint distortion is large and the tax rate is high.

(ii) **Inverse hazard rate** ($1/\alpha$)**.** The Pareto inverse hazard rate (49) is the ratio $(1-F)/(zf)$ that converts the magnitude of λ per person into the tax wedge per unit of income. With a thicker tail (smaller α), more types are affected by any given marginal rate, so the revenue gain from a rate increase is large relative to the behavioral distortion at the margin.

(iii) **Elasticity** (e)**.** The factor $(1+\varepsilon^u)/\varepsilon^c = (1+\gamma)$ multiplies the inverse hazard rate. A more elastic labor supply (smaller $1/\gamma$, i.e., larger e) amplifies the behavioral revenue loss from a rate increase, pushing the optimal rate down.

The HJB perspective makes these three forces transparent: they are the gradient of the value function (λ), the geometry of the type distribution (f and F), and the curvature of the flow payoff (the elasticity terms in the maximand).

8.7 Zero top rate under bounded support vs. positive top rate under Pareto: a unified view

In both cases the tax formula is $T'/(1-T') = -(1+\varepsilon^u)/\varepsilon^c \cdot \lambda(z)/(zf(z)\mu)$. The difference is entirely in the behavior of $\lambda(z)/[zf(z)]$:

- **Bounded support:** $1 - F(z) \to 0$ as $z \to \bar{z}$ faster than density $f(z) > 0$, so $(1 - F(z))/(z\, f(z)) \to 0$. Combined with $\lambda(\bar{z}) = 0$, the tax formula gives $T'(\bar{z}) = 0$.

- **Pareto tail:** $(1 - F(z))/(z\, f(z)) \to 1/\alpha > 0$ and $\lambda(z) \approx -(1 - F(z))$, so the ratio $\lambda(z)/(zf(z)) \to -1/\alpha \neq 0$. The tax formula gives $T'_\infty = 1/(1 + \alpha_y e) > 0$.

The zero-rate result and the Diamond–Saez formula are thus two limiting cases of the same HJB first-order condition, differing only in the asymptotic behavior of the Pareto inverse hazard rate.

8.8 Asymptotic value function under Pareto

Under the linear guess the value function is $V(u, z; \mu) = \lambda(z)\, u + K(z; \mu)$. For large z:

$$\lambda(z) \approx -(z_0/z)^\alpha, \qquad K(z; \mu) \approx -\int_z^\infty \mathcal{F}(\zeta)\, d\zeta \tag{57}$$

where $\mathcal{F}(\zeta) = \mu[y^*(\zeta) - C(\mathcal{U}^*(\zeta), y^*(\zeta))] f(\zeta)$ is the flow resource surplus at ζ. Both $\lambda(z)$ and $K(z; \mu)$ decay polynomially at rate $z^{-\alpha}$, consistent with the transversality condition $V(u, z; \mu) \to 0$. The rate of decay is the Pareto shape α: heavier tails (smaller α) slow the decay of the continuation value, put more weight on incentive constraints at high types, and push up the optimal tax rate.

8.9 Table: bounded support vs. Pareto tail

Remark 9 (Empirical calibration)**.** *The Diamond–Saez formula* (56) *requires two empirical inputs: α_y (the Pareto shape of the earnings distribution) and e (the compensated elasticity of taxable income). Saez (2001) surveys estimates suggesting $\alpha_y \approx 2$ for the United States, and a benchmark value of $e \approx 0.25$ for broad income, giving $T'_\infty \approx 1/(1 + 2 \times 0.25) = 2/3 \approx 67\%$. With higher elasticity, say $e = 0.5$, the formula gives $T'_\infty = 1/(1+1) = 50\%$. Diamond and Saez (2011) conclude that T'_∞ in the range 50–70% is consistent with the empirical evidence, well above the top marginal rates prevailing in most countries at the time.*

Table 2: Mirrlees HJB: Bounded vs. Pareto-tailed productivity

Bounded support $\mathcal{Z} = [z_0, \bar{z}]$	**Pareto tail** $\mathcal{Z} = [z_0, \infty)$
Upper BC: $\lambda(\bar{z}) = 0$ (point condition)	Upper BC: $\lambda(\infty) = 0$ (transversality)
$(1 - F(\bar{z}))/(z_{\bar{z}} f(\bar{z})) = 0$	$(1 - F(z))/(zf(z)) \to 1/\alpha > 0$
$\lambda(z)/[zf(z)] \to 0$ as $z \to \bar{z}$	$\lambda(z)/[zf(z)] \to -1/\alpha < 0$
Zero top rate: $T'(\bar{z}) = 0$	Positive top rate: $T'_\infty = 1/(1 + \alpha_y e) > 0$
Value fn. decay: $V(u, z) \to 0$ at finite $\bar{z}$	Value fn. decay: $V(u, z) \sim z^{-\alpha} \to 0$
Costate: $\lambda(z) \geq 0$ is hump-shaped	Costate: $\lambda(z)$ turns negative for large z

A The Hamiltonian and HJB for Mirrlees: Equivalence

The Hamiltonian approach of the lecture notes defines

$$\mathcal{H}(\mathcal{U}, y, \lambda;\, z) = \big[\psi(z)\,\mathcal{U} + \mu(y - C(\mathcal{U}, y))\big] f(z) - \lambda\, U_n\big(C(\mathcal{U}, y), y/z\big)\frac{y}{z^2}. \tag{58}$$

The HJB (25) is the Hamilton–Jacobi equation

$$-V_z(u, z) = \mathcal{H}\big(u,\, y^*(u, z),\, V_u(u, z);\, z\big), \tag{59}$$

where $y^*(u, z) = \arg\max_y \mathcal{H}(u, y, V_u(u, z); z)$. Along the optimal path, the costate $\lambda(z) = V_u(\mathcal{U}^*(z), z)$ satisfies $d\lambda/dz = -\mathcal{H}_{\mathcal{U}}$ (equation (37) above), confirming that the Hamiltonian and HJB approaches are equivalent.